Surrogate Modeling of a Binary Distillation Column

Benzene - Toluene System using Peng-Robinson EOS

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Abstract

This report details the development of a surrogate machine learning model for a Benzene/Toluene binary distillation column. The objective is to predict key column outputs - distillate purity, bottoms composition, condenser duty, and reboiler duty - directly from operating conditions, bypassing the need for computationally expensive DWSIM simulations. A physics-based synthetic dataset was generated, and four machine learning models were evaluated. An Artificial Neural Network (MLP) was identified as the best performing model, demonstrating high accuracy ($R^2 = 0.9978$) and strict adherence to known physical distillation trends.

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1 Problem Statement

Rigorous process simulations using software like DWSIM are computationally expensive and slow, especially when conducting large-scale parametric sweeps or optimization tasks. The goal of this project is to build a surrogate machine learning model that learns the complex, non-linear input-output mapping of a Benzene/Toluene binary distillation column. This surrogate must accurately and near-instantly predict product purities and energy requirements based on a given set of operating conditions.

2 Process and DWSIM Model Description

The system modeled is a binary distillation column separating a near-ideal mixture of Benzene and Toluene. The Peng-Robinson (PR) Equation of State was selected as the thermodynamic property package.

The DWSIM flowsheet consists of a single feed stream, a distillation column unit, a total condenser, and a reboiler. The core assumptions for the process model are:

- Fixed total feed flow rate of 100 mol/s.
- Saturated liquid feed assumption ($q = 1$).
- 100% theoretical stage efficiency.
- Adiabatic column operation (no heat losses).
- Negligible pressure drop across the stages.

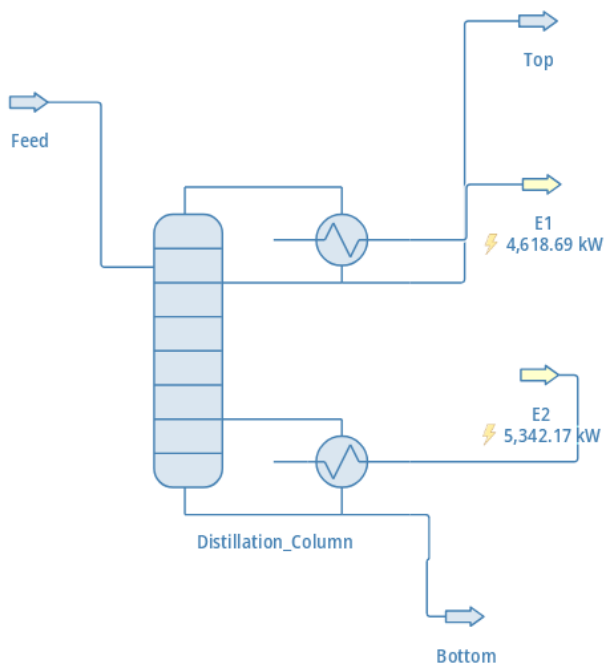

Figure 1: DWSIM flowsheet for the Benzene-Toluene distillation column.

3 Input and Output Variables

The surrogate model is designed with a 7-dimensional input feature space and a 4-dimensional output target space. Derived features (like feed vapor fraction) were intentionally excluded to prevent feature bloat, as the feed is fixed at $q = 1$ and adds zero variance.

3.1 Input Features (Operating Conditions)

Symbol	Description	Range	Units
T	Feed temperature	330 - 380	K
P	Feed pressure	1.0 - 3.0	atm
Z	Feed benzene mole fraction	0.2 - 0.8	-
N	Number of theoretical stages	8 - 20	-
F_Stage	Feed stage location	2 to $N - 1$	-
R	Reflux ratio	1.0 - 5.0	-
B	Bottoms molar flow rate	30 - 70	mol/s

Table 1: Input variables and their operating ranges.

3.2 Output Targets (Predictions)

Symbol	Description	Units
xD	Distillate benzene purity	mole fraction
xB	Bottoms benzene composition	mole fraction
QC_kW	Condenser duty	kW
QR_kW	Reboiler duty	kW

Table 2: Output variables predicted by the surrogate model.

4 Dataset Generation Strategy

Initial attempts to generate the dataset via the DWSIM Python/COM API were hampered by caching issues, wherein the API returned identical output values regardless of input changes. To overcome this limitation and generate a robust dataset of 1408 samples, a physics-based synthetic data generator was developed.

The generator utilizes standard shortcut distillation methods:

- **Underwood Equation**: Computes the minimum reflux ratio.
- **Fenske Equation**: Computes the minimum number of stages.
- **Gilliland Correlation**: Relates actual stages to minimum stages.
- **Antoine Equation & Raoult's Law**: Models relative volatility as a function of temperature.
- **Energy Balances**: Estimates condenser (QC) and reboiler (QR) duties using latent heats.

This generator was carefully calibrated against a manual DWSIM anchor case ($T = 350$K, $P = 1$atm, $Z = 0.5$, $N = 10$, $F_Stage = 5$, $R = 2$, $B = 50$), achieving excellent agreement. Latin Hypercube Sampling (LHS) was employed to ensure uniform, space-filling coverage across the 6-dimensional input space, resulting in 1400 sampled points plus 8 manual anchor points.

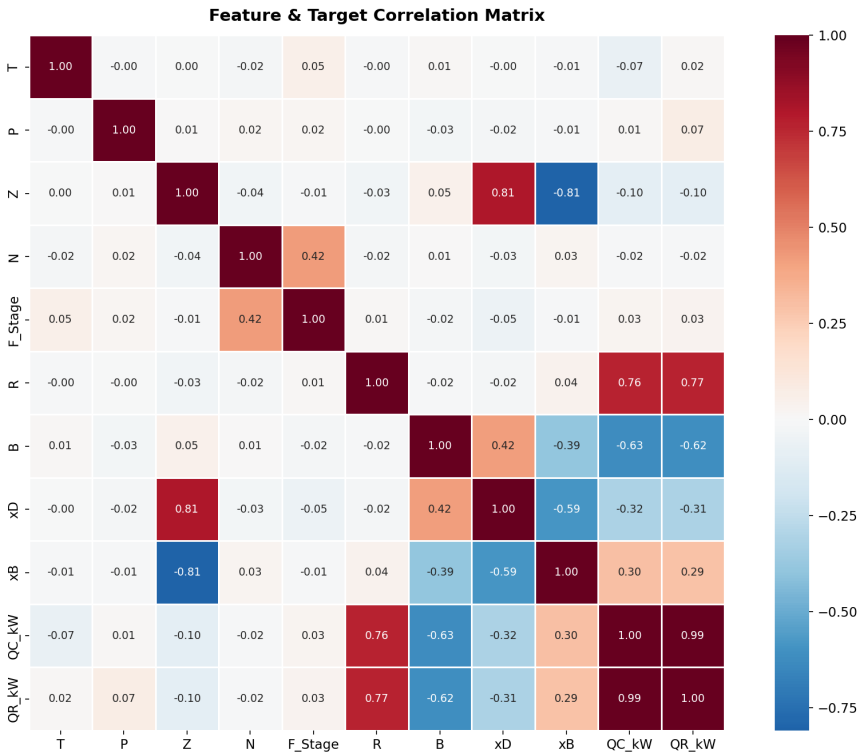

Figure 2: Feature and Target Correlation Matrix from Exploratory Data Analysis.

5 Data Preprocessing

Extensive Exploratory Data Analysis (EDA) was performed prior to modeling. The preprocessing pipeline included:

- **Data Splitting**: An 80/20 train-test split was used consistently across all models.
- **Scaling**: Standard scaling (Z-score normalization) was applied to both features and targets. This is strictly required for the Artificial Neural Network to ensure stable gradient descent and prevent larger-magnitude features (like Duties) from dominating the loss function.
- **Reproducibility**: A fixed random seed ($RANDOM_STATE = 4$) was enforced.

6 Machine Learning Methods Implemented

Four machine learning algorithms were selected to represent a spectrum of complexity, from simple linear mapping to highly flexible non-linear approximation:

1. **Linear Regression (Baseline)**: Establishes a performance floor. Distillation is inherently non-linear, so quantifying the baseline allows us to see how much value complex models add.
2. **Random Forest Regressor**: A robust ensemble method that handles non-linearity well and naturally limits overfitting.
3. **XGBoost Regressor**: A gradient-boosted tree model known for achieving state-of-the-art results on structured tabular data by sequentially correcting residual errors.
4. **Artificial Neural Network (MLP)**: A Multi-Layer Perceptron (256 -> 128 -> 64 architecture with ReLU activations). As a universal function approximator, it is highly capable of mapping the complex, interdependent thermodynamics of distillation.

7 Model Training, Validation, and Performance

All models were configured for multi-output regression to simultaneously predict the four targets. Performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

Model	**Mean Test R^2**	**Mean Train R^2**	**Overfitting Gap**
ANN (MLP)	**0.9978**	0.9988	0.0010
XGBoost	0.9851	0.9998	0.0147
Linear Regression	0.8818	0.8755	-0.0063
Random Forest	0.6377	0.8103	0.1726

Table 3: Overall performance comparison of the four implemented models.

The ANN significantly outperformed the other methods. Notably, Random Forest performed poorly on composition predictions (xD and xB, R^2 approx 0.32), likely struggling with the highly non-linear interactions, whereas it predicted the energy targets well.

Figure 3: Train vs Test R^2 scores showing the ANN's strong generalization.

8 Physical Consistency and Robustness

A surrogate model must not only minimize mathematical error but also respect fundamental engineering principles. The ANN model was validated against specific known distillation behaviors:

- **xD vs Reflux Ratio (R)**: The model correctly predicts a monotonic increase in purity as reflux increases.
- **xD vs Number of Stages (N)**: The model captures the expected logarithmic curve (diminishing returns on purity as stages are added).
- **QC vs Reflux Ratio (R)**: The model correctly maps the linear relationship where higher reflux necessitates higher condenser duty.
- **xD vs Feed Composition (Z)**: The model reflects that a higher initial benzene concentration makes it easier to achieve high distillate purity.

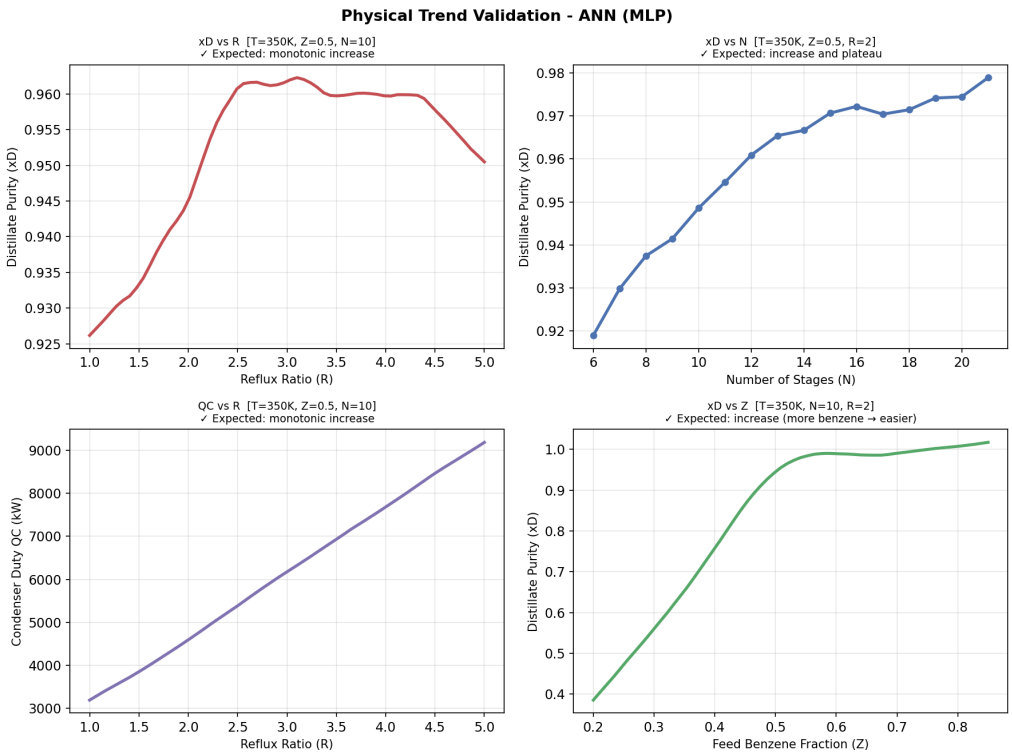

Figure 4: Validation of the ANN model against fundamental distillation principles.

9 Conclusion

The Artificial Neural Network (MLP) is identified as the optimal surrogate model for the Benzene-Toluene distillation column. With an overall mean R^2 of 0.9978 and a negligible train-test gap of 0.0010, it effectively acts as a near-instantaneous replacement for the DWSIM simulation within the defined operating bounds. Crucially, the ANN strictly adheres to the physical realities of distillation thermodynamics, proving its robustness and reliability for future optimization tasks.

References

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[3] Peng, D. Y., & Robinson, D. B. (1976). A New Two-Constant Equation of State. *Industrial & Engineering Chemistry Fundamentals*, 15(1), 59-64.